Data Verification - AMF1 Partner Survey Dashboard

1. Purpose & Context

One of the most important disciplines in any data product is the ability to verify that what the platform shows is actually correct. For a dashboard built on survey data — where clients make commercial decisions based on what they see — this is not optional. It has to be systematic, repeatable, and evidence-backed.

This brief is to build the data integrity and testing infrastructure for the AMF1 Partner Survey Dashboard. You are not being handed a pre-built approach. You are being asked to design and build it, using the tools, access, and context available to you. The outputs will be used directly by the product team and will set the standard for how we approach this across future projects.

2. Project Context

The AMF1 Partner Survey Dashboard is a data visualisation platform built for Aston Martin F1's research team. It surfaces brand tracking survey data across 7 markets and 18 partner brands, enabling wave-over-wave comparison and demographic segmentation.

Key Detail	Description
Client	Matt Roberts (Potentia-Sports) for AMF1 (Aston Martin F1)
Markets	UK, USA, Japan, Mexico, Australia, Saudi Arabia, China
Partner Brands	18 brands — Overview page plus 14 dedicated partner pages
Waves	Wave 1 and Wave 2 — both ingested and live in the sandbox environment
Respondents	~13,000 per wave (~26,000 total across both waves)
Environment	Sandbox — development build, not production

The PRD is your reference for features, navigation, and chart types. It is approximately 90% accurate. Treat it as the baseline. If you find discrepancies between the PRD and what the dashboard actually shows, log them as findings rather than assuming either is wrong — Suraen will triage.

3. Your Access

You will have the following from day one:

- Live sandbox dashboard URL
- Global tables — multiple Excel files covering Wave 1 and Wave 2, broken down by market and individual filter options. These are your primary source of truth.
- Wave 1 metadata CSV — variable-to-label mappings for all question codes
- QA reference workbook — available via the shared Google Drive folder
- Codebase and repository access

4. How to Approach This

You have flexibility in tooling. What matters is that the outputs are accurate, reproducible, and readable by someone who wasn't there when you built them. The architecture for the data verification layer is covered in Phase 1.

On using AI as a working partner

You are encouraged to use AI tools throughout this project to move faster and validate your thinking. One specific recommendation given we are on Google Workspace: set up a NotebookLM project as part of your Phase 0 orientation. Load the PRD, your reference document, and any other project files into it. This gives you a project-aware AI assistant that can help you reason through data model questions, sense-check your logic, and pressure-test your reference document before you start comparing values against it.

The distinction that matters: use AI to build the tool, not as the tool itself. The verification script should query the API directly and compare values programmatically — no AI model in the runtime loop. Keep the extraction and comparison layers deterministic.

On reusability

Everything you build here should be architected for reuse — not just for this survey, but across other AMF1 projects and beyond. This means the reference document format, the API query layer, and the comparison logic should treat the survey structure, question definitions, and API endpoints as configurable inputs rather than hardcoded values. Build an MVP, but build it so that pointing it at a different survey or a different client requires configuration, not a rewrite. This is a meaningful constraint: keep it in mind from the first line of code you write.

5. Work Phases

Your work runs across four phases. Phase 0 must be completed before Phase 1 begins. Phases 1 and 2 can overlap once your reference document is stable. Phase 3 is lowest priority and should only begin once Phases 1 and 2 are well advanced.

Phase 0 — Orientation & Reference Document

Before writing a single line of test code, you need to deeply understand the data. This phase is deliberately a learning exercise. The act of building the reference document will give you the domain context you need to write meaningful tests.

What to do

- Book a 30-minute onboarding call with Suraen. The most important things to understand are: the survey routing chain (how D1 feeds D2 feeds D3/D4/D5, and why the sample base changes per question), how the 18 partner brands relate to the 31 awareness brands tracked at D1, and how wave-over-wave comparison works in the platform.
- Read the PRD in full. Map every question in scope to its chart type, base population, and filter behaviour. Note anything that is unclear or ambiguous.
- Open the global table files for Wave 1 and Wave 2. These are pre-processed summary outputs generated by the survey company. They contain independently computed percentage values for each question, broken down by market and individual demographic filter options. This is your source of truth — not the PRD, not the QA reference workbook.
- Set up a NotebookLM project and load the PRD and your reference document into it. Use it as a working assistant throughout the project for data model questions and logic checks.
- Build a structured reference document — Excel or structured CSV — that captures expected values for every in-scope question at total sample level and for each individual filter slice. This

document will govern Phase 1.

Expected outputs

- Reference document covering all in-scope questions with expected values at total sample and per individual filter option
- Short written summary of anything in the global tables you found ambiguous or could not cleanly map to the dashboard — share with Suraen before Phase 1 begins

Phase 1 — Data Verification (Primary Priority)

This is the most important phase. The goal is to build an automated verification tool that confirms the data surfaced in the dashboard matches your reference document — for every in-scope question, on every page, for every individual filter slice. Coverage target: 100%.

Scope

Verification covers the Overview page and all 14 partner pages. For the Overview page, the P0 questions are D1–D6 (brand awareness, familiarity, usage, consideration, favourability, NPS, and sponsor awareness). For all partner pages, the full set of partner-specific questions as documented in the PRD applies at P1. Refer to the PRD for the complete question list per partner page — these are not repeated here to avoid duplication.

Area	Scope	Priority
Overview page	D1 Brand Awareness (note: SUM of D1_1r + D1_2r per brand), D2 Familiarity, D3a Currently own/use, D3b Consider using, D4 Favourability, D5 NPS (Promoters 9–10 minus Detractors 0–6), D6 Sponsor Awareness	P0

All 14 partner pages	Full question set per partner as documented in the PRD	P1

Architecture

The verification tool works in three layers:

- Reference layer — your reference document (from Phase 0) is parsed into a structured dataset. This is the ground truth the tool compares against.
- Query layer — the tool calls the Cube.js API with each question and filter combination, capturing the JSON response. No browser automation, no screenshot interpretation, no AI model in the loop. Direct API calls returning structured data.
- Comparison layer — the tool diffs API response values against the reference dataset, flags mismatches, and records the delta.

> *Before building the query layer, discuss the Cube.js query structure with Kasun. You need to understand how filters are passed as parameters in order to construct the right API call per question and filter option. Validate your proposed approach with him before implementing — he may have a simpler or smarter way to do this. Do not spend more than a day building before getting that input.*

Filter scope

Verify at two levels for every in-scope question:

- Total sample — default state with no filters applied
- Individual filter slices — each filter option applied in isolation (e.g. UK only, Male only, F1 Fans only, each team fan segment, etc.)

Filter combinations (e.g. UK + Male + F1 Fan simultaneously) are explicitly out of scope for this phase. The global tables do not contain combination-level breakdowns, so there is no ground

truth to compare against. This is a separate workstream to be scoped with Sekar (CTO). Flag anything you learn during Phase 1 that would inform how that could be done at scale.

Expected outputs

- Automated verification script — documented, re-runnable with a single command, configurable for reuse across other surveys
- Pass/fail report for every question and filter slice tested, with delta values where mismatches are found
- Prioritised log of data discrepancies with enough detail for the dev team to investigate

Phase 2 — Functional Testing (Secondary Priority)

Once data verification is underway, build a functional test suite that validates the behaviour of the dashboard independently of whether the numbers are correct.

Coverage areas

- **Filters & Interactions:** All filter options render correctly
  - Selecting a filter updates charts without breaking layout
  - Switching between markets, demographics, and fan segments behaves as expected
  - Resetting filters returns the dashboard to its default state

- **Wave Switching:** Toggling between Wave 1 and Wave 2 updates all charts correctly
  - Delta indicators (percentage point change) display and are directionally correct
  - Wave 2-only questions and filters (e.g. Tech Adopters at D17a) render correctly in a Wave 2 context and show N/A or are appropriately hidden in a Wave 1 context
  - Wave 1-only questions behave symmetrically — verify correct rendering and appropriate fallback or N/A states when viewed in a Wave 2 context
  - Error and fallback states are tested explicitly — document what the dashboard shows when data is absent for a given wave, question, or filter combination

- **Partner Pages:** Partner pages load and display correct brand-specific content
  - N/A states render correctly for questions not applicable to a given brand
  - Glenfiddich-specific row in D14 appears only for Glenfiddich

- **Chart Rendering:** Charts render without visual errors across page types
  - Low-n warnings appear when sample size falls below 50
  - Sample sizes update correctly when filters are applied

Expected outputs

- Structured test case log covering all areas above — can be manual or automated
- Bug and behaviour log with screenshots or recordings for any issues found

Phase 3 — Performance Testing (Tertiary Priority)

Start this only once Phases 1 and 2 are well advanced. The goal is to establish baseline load and response benchmarks for the sandbox environment.

Coverage areas

- Initial page load time for the Overview dashboard and a representative partner page
- Filter response time — time from filter selection to chart update
- Wave switching response time
- Behaviour under multiple concurrent users — basic load scenario

Expected outputs

- Benchmark report with load times for key interactions
- Any performance bottlenecks identified with enough detail for the dev team to investigate

6. Definition of Done

Phase	Completion Criteria
Phase 0	Reference document built from global tables, ambiguities resolved with Suraen, ready to use as Phase 1 ground truth. NotebookLM project set up with relevant files loaded.
Phase 1	Automated verification tool running. 100% of in-scope questions verified at total sample and per individual filter slice across Overview and all partner pages. Pass/fail report produced. Discrepancies logged and shared.
Phase 2	Functional test cases completed across all coverage areas including wave-specific states and error/fallback scenarios. Bug log up to date with screenshots.
Phase 3	Benchmark report produced with key load and response metrics. (Time permitting.)
Overall	All outputs documented, reproducible, and transferable to another team member without requiring explanation from you. Verification tool is configurable for reuse on other surveys without a rewrite.

7. Working Norms & Check-In Cadence

- Daily async Slack update: what you tested, what you found, any blockers.
- Do not spend more than 30 minutes stuck on a data interpretation or access question before flagging it. Escalate early.
- Data discrepancies should be logged with: the specific question, the filter applied, the expected value from your reference document, and the actual value from the dashboard. Suraen triages.

- Propose your tooling and query approach before implementing. A short Slack message is sufficient. Do not build first and ask later.
- Technical questions about the Cube.js API and ETL pipeline go to Kasun. All other questions go to Suraen first.

> *Your goal is not to find bugs — it's to give the team a clear, evidence-backed picture of where the dashboard stands. A well-structured 'all passing' report is just as valuable as a list of failures.*

8. Out of Scope

- D13 Word Association — separate workstream, not part of this brief
- Filter combination verification — requires microdata access, separate workstream to be scoped with Sekar (CTO)
- Wave 3 data — not yet received
- Fan Survey dashboard — separate product
- Data export functionality
- User access controls

9. Key Resources

Resource	Notes
PRD: AMF1 Partner Survey Dashboard	Features, navigation, chart types, filter definitions. 90% accurate — log discrepancies, don't assume errors.
Global Tables (Wave 1 & Wave 2)	Multiple Excel files. Your primary source of truth for Phase 1. Shared via project folder.
Wave 1 Metadata CSV	Variable-to-label mappings. Use to decode question codes (e.g. D1_1r, D2, etc.).

QA Reference Workbook	Available via Google Drive. Useful secondary reference — do not treat as the ground truth for data values.
NotebookLM	Set up your own project notebook as part of Phase 0 orientation. Load the PRD and your reference document. Use as a working assistant for data model and logic questions throughout the project.
Sandbox Dashboard URL	Shared by Suraen at onboarding.
Codebase / Repository	Access via [repo link]. Review the Cube.js API layer to understand query structure before building the verification tool.
Suraen (PM)	Daily Slack. First point of contact for all product and data questions.
Kasun (Dev Lead)	Escalations on Cube.js API structure, ETL pipeline, and database schema.
Sekar (CTO)	Future conversation on filter combination verification at scale — not in scope for this brief.